

TASK 5: EXPLORATORY DATA ANALYSIS (EDA)

INN Hotels Group — Hotel Booking Dataset

Detailed EDA Report

Objective: Explore booking patterns, customer characteristics, room pricing and cancellation behaviour using statistical and visual analysis.

Prepared from the completed Google Colab notebook: Task_5 (5).ipynb

1. Problem Definition and Objective

INN Hotels Group has a chain of hotels in Portugal and is facing a high number of booking cancellations. The purpose of this exploratory analysis is to understand the structure of the booking data, identify important patterns and relationships, and examine factors associated with cancellation behaviour. The findings can support future predictive modelling and help the business develop more informed cancellation and refund policies.

The EDA covers statistical summaries, distributions of numerical variables, categorical frequency analysis, bivariate relationships, multivariate relationships, arrival-month patterns, missing-value checks and outlier analysis.

2. Dataset Overview

Item	Finding
Number of records	36,275
Number of variables	19
Data types	Numerical and categorical variables
Missing values	No missing values reported
Main outcome examined	Booking status / cancellation

The dataset contains booking-level information such as lead time, number of adults and children, length of stay, average price per room, market segment, room type, repeated-guest status, previous cancellations and special requests. Categorical variables include Booking_ID, type_of_meal_plan, room_type_reserved, market_segment_type and booking_status.

3. Data Quality and Statistical Exploration

The notebook reports no missing values, so no missing-value imputation was required for the EDA. Duplicate checking and descriptive statistics were performed before visual analysis. Numerical variables were examined through measures of central tendency and spread, while categorical variables were examined through frequency distributions.

Average room price is right-skewed and contains high-value observations. Some bookings have an average room price of zero, which is consistent with complimentary bookings identified in the market-segment analysis. Lead time is also right-skewed, indicating that most bookings have relatively shorter lead times while some are made much further in advance.

4. Univariate Analysis

Guest composition: Most bookings are for two adults. There are also a significant number of bookings for one adult, while larger adult groups occur less frequently. The number of children has a relatively small distribution, with a few unusual high observations.

Stay duration: Weekend-night bookings are concentrated around shorter stays. The distribution is skewed toward shorter weekend stays, with one- and two-night bookings occurring frequently.

Market segment: The Online segment contributes the largest number of bookings by a significant margin, followed by Offline. Corporate contributes a smaller share, while Complementary and Aviation have much lower

booking volumes.

Special requests: The majority of bookings have no special requests. A substantial number have one special request, while two or more special requests are less common.

5. Bivariate and Multivariate Analysis

Lead time and booking status: The analysis shows a positive association between lead time and cancellation, suggesting that bookings made further in advance are more likely to be canceled in this dataset.

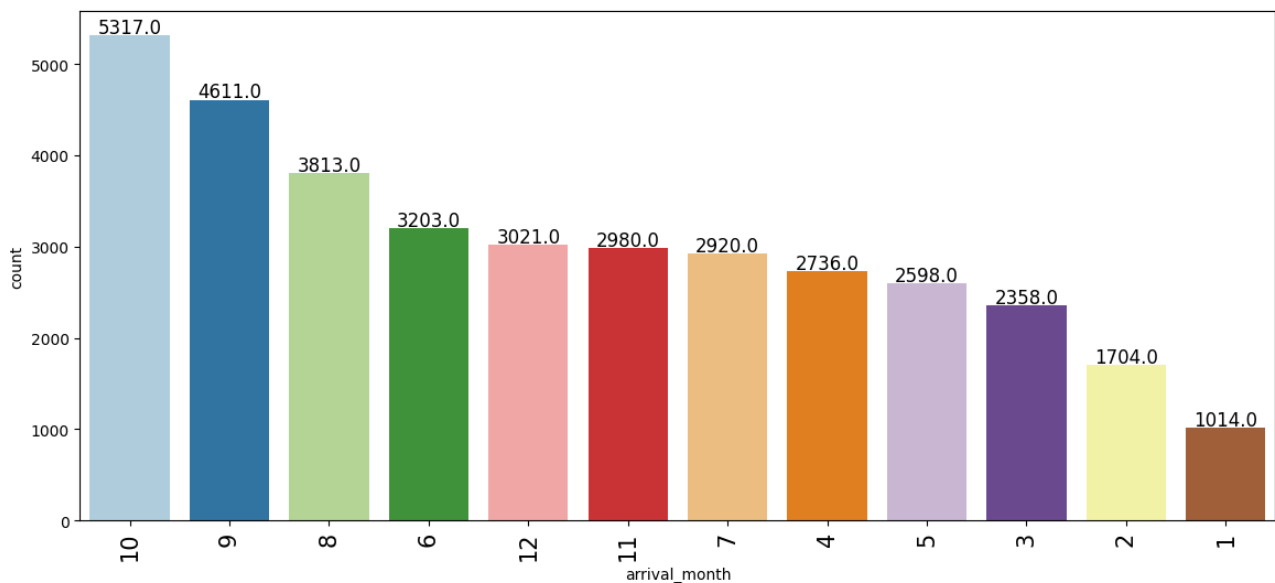
Special requests and booking status: Bookings with no special requests have the highest cancellation proportion. As the number of special requests increases, the cancellation proportion generally decreases. Bookings with three or more special requests have no cancellations in this dataset. This indicates a negative association, but does not establish causation.

Room price and market segment: Online and Offline segments show a wide range of room prices with relatively similar medians. Corporate generally has lower median prices and a smaller spread, while Complementary bookings include zero-priced rooms. Aviation shows a relatively higher median price.

Correlation structure: The heatmap highlights relationships among numerical variables, including the positive association between lead time and booking cancellation and the negative association between special requests and cancellation.

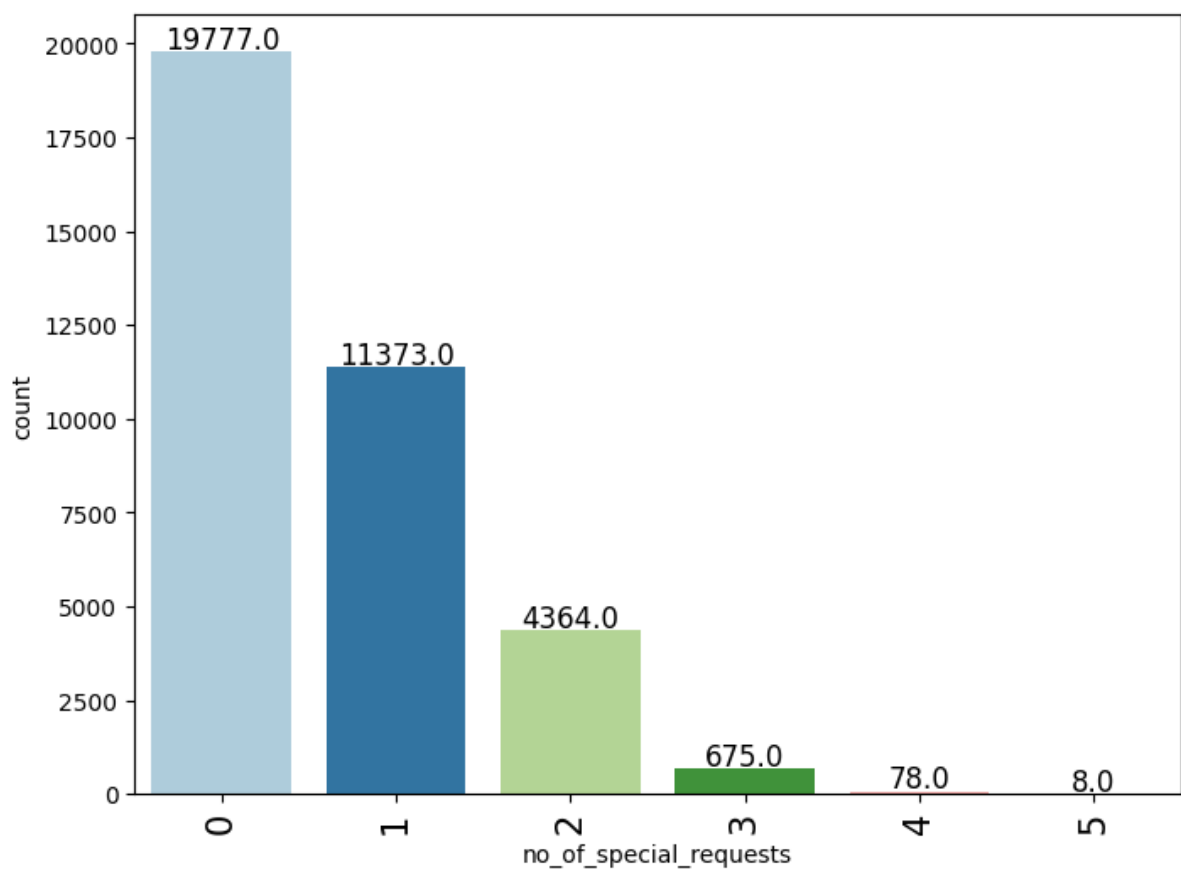
6. Key Visual Evidence

Booking volume by arrival month



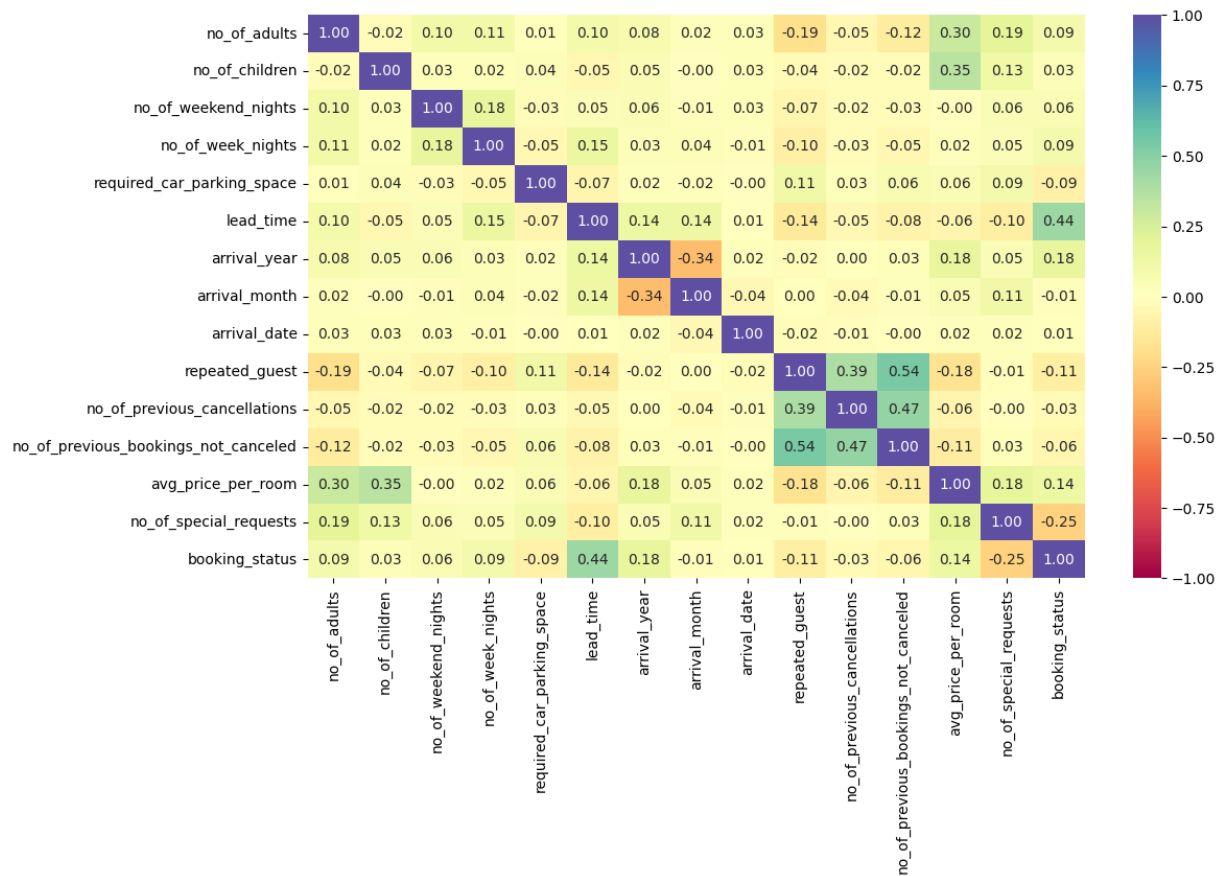
The arrival-month analysis shows the highest booking volume in October, followed by September and August.

Special requests and booking status



The cancellation proportion generally decreases as the number of special requests increases.

Correlation heatmap



The heatmap summarizes correlations among numerical variables and supports the relationships discussed in the bivariate analysis.

7. EDA Questions and Detailed Findings

Q1. What are the busiest months in the hotel?

October has the highest number of bookings, followed by September and August. The monthly pattern indicates that booking activity is considerably higher in the later part of the year than in January and February.

Q2. Which market segment do most of the guests come from?

The Online market segment contributes the largest number of guests by a significant margin. Offline is the second-largest segment, while Corporate contributes a smaller share and Complementary/Aviation account for relatively few bookings.

Q3. What percentage of bookings are canceled?

The notebook reports approximately 32.76% of bookings as canceled. This makes cancellation a substantial business issue and an important outcome for further modelling.

Q4. What percentage of repeating guests cancel?

Approximately 1.72% of repeating guests cancel. The substantially lower rate compared with the overall cancellation rate suggests that repeat customers are considerably less likely to cancel in this dataset.

Q5. Do special requirements affect booking cancellation?

Bookings with no special requests have the highest cancellation proportion. As the number of special requests increases, the cancellation proportion generally decreases. This is an association observed in the dataset and should not be interpreted as proof that special requests cause lower cancellation.

Q6. What are the differences in room prices in different market segments?

Online and Offline segments have a wide range of average room prices with relatively similar median prices. Corporate has a lower median price and smaller spread, which may reflect negotiated corporate rates. Complementary bookings have a price of zero, while Aviation has a relatively higher median price.

8. Outlier Analysis

Boxplots identify outlying observations in several numerical variables, particularly average room price. The analysis does not automatically remove these observations because extreme values may represent genuine differences in room pricing, market segments or booking conditions. Complimentary bookings also contribute to the lower end of the price distribution.

The final notebook decision is to retain the identified outliers for EDA rather than treating them as data errors. This preserves potentially meaningful business information for subsequent modelling or investigation.

9. Consolidated Findings

1. October, September and August have the highest booking volumes.
2. Online is the largest market segment.
3. Approximately 32.76% of bookings are canceled.
4. Only approximately 1.72% of repeating guests cancel.
5. Bookings with special requests generally show lower cancellation proportions.

6. Lead time is right-skewed and has a positive association with cancellation.
7. Average room price is right-skewed and contains genuine extreme values as well as complimentary zero-priced bookings.
8. Room pricing varies across market segments, with Corporate generally lower and Complementary at zero.
9. The analysis finds associations and patterns; these EDA results do not by themselves establish causal relationships.

10. Conclusion

The exploratory analysis provides a detailed view of booking behaviour in the INN Hotels dataset. Cancellation is substantial overall but is much lower among repeating guests. Booking lead time, market segment, special requests and room pricing show meaningful relationships with booking behaviour. Seasonal booking volume is also evident, with October showing the highest number of bookings. These findings provide a strong foundation for subsequent predictive modelling and for designing data-driven cancellation and refund policies.

The EDA should be treated as an exploratory stage: the observed relationships can guide feature selection and business hypotheses, while predictive modelling and statistical testing would be required for stronger conclusions.

Source: Completed Google Colab notebook — Task_5 (5).ipynb